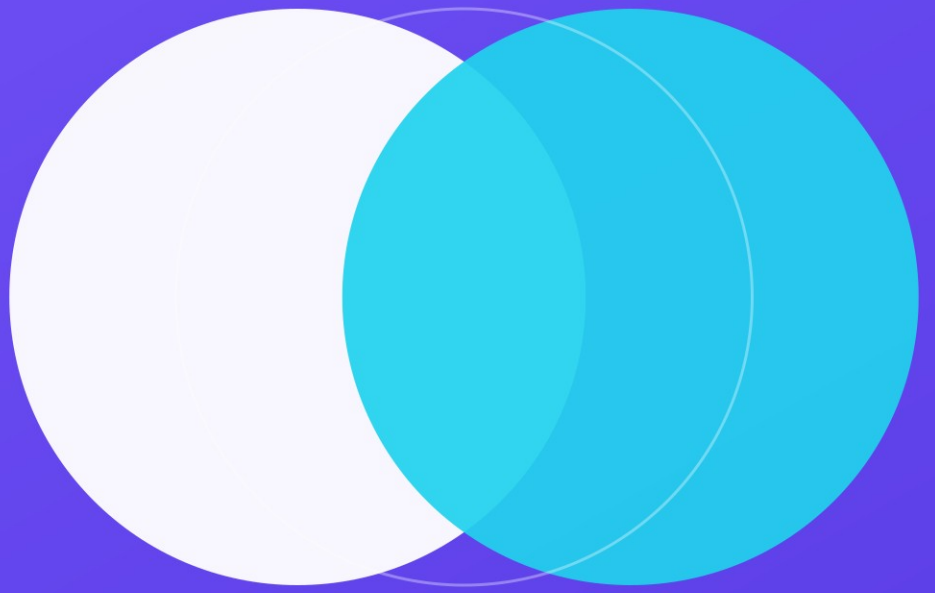
CREATOR AI · LIGHTNING TRACK · ASUS UGEN300

SceneGap Edge

Know which must-have shots you already have, and which to reshoot, before the crew leaves the location.



Local-first

Evidence for every verdict

Person confirms

Hailo-10H route

Stage I proposal · [Team name] · github.com/[team]/scenegap-edge

PROBLEM

The missing shot is found at 23:00, after the café has closed

Two-person crew, café promo, 11 must-have shots

“We had pours. None of them was three seconds long. We found out in the edit.”

09:00	Shot list agreed with the client: storefront, interior, beans, grinder, 3 s pour, two barista takes, bag with readable brand.
17:40	Wrap. Several cards of footage; nobody has time to watch it all on site.
23:00	The editor scrubs for the pour: every take cuts away after 2 s. The brand on the bag is out of focus.
Next week	Reshoot day: crew, café, client sign-off again.

PROBLEM

Editors can now search footage; nobody checks the shot list

TOOL	SEARCHES FOOTAGE	TIME RANGE IN CLIP	CHECKS DURATION / TAKES / SAME FRAME	RUNS LOCALLY	PRICE
Premiere Pro Media Intelligence	yes	not stated	no	yes	~$22.99/mo (secondary)
DaVinci Resolve 21 IntelliSearch	yes	visual: whole clips	no	likely	Studio $295
Final Cut Pro 12 Visual Search	yes	yes	no	not stated	$299.99
Frame.io semantic search	yes (beta)	yes	no	no (cloud)	$25/member/mo
Jumper	yes	yes	no	yes	from $49/mo
invideo AI script supervisor	vs shot list	not stated	done/pending only	upload	not listed
SceneGap Edge	**yes**	**yes**	**yes, with 'unknown'**	**yes**	**see business**

> **Search tells you where a cup appears. SceneGap tells you whether the 3-second pour you promised exists.**

Vendor pages and trade press, accessed 2026-10-26; full list with URLs in docs/research/BUSINESS_FACTS.md.

SOLUTION

Requirement → constraint → evidence → decision

01

Write the shot list

In Chinese, as the crew speaks. A person confirms the English query for the model and adds rules: min. length, takes, same-frame.

Host CPU

02

Index footage locally

Probe, hash, cut shots, sample 1 frame/s on the original timestamps. CLIP image embeddings.

UGen300

03

Check each rule

Runs of matching samples inside one shot; conservative duration bounds; 250 ms re-sampling when undecided.

Host CPU

04

Decide with evidence

Candidate / partial / not found / unverifiable, each with playable clips. A person accepts; gaps become reshoot cards.

Host CPU

UGen300 box = planned route A; today the same model runs on the host CPU/GPU and is labelled cpu-/cuda-reference.

SOLUTION

Four answers per shot, and a person has the last word

The AI's answer and the person's answer are stored separately; only the person's counts as done.

Candidate

Every rule passes on at least one segment. Still needs a person to accept.

Partial

Something matched but a rule failed or is undecided: too short, one take of two, a manual check left.

Not found

Nothing reached the threshold in the analysed scope. A reshoot card is proposed, with the nearest frames to check first.

Unverifiable

Files failed to decode or are missing, so absence cannot be claimed. Fix the scope first.

Human: accepted / rejected

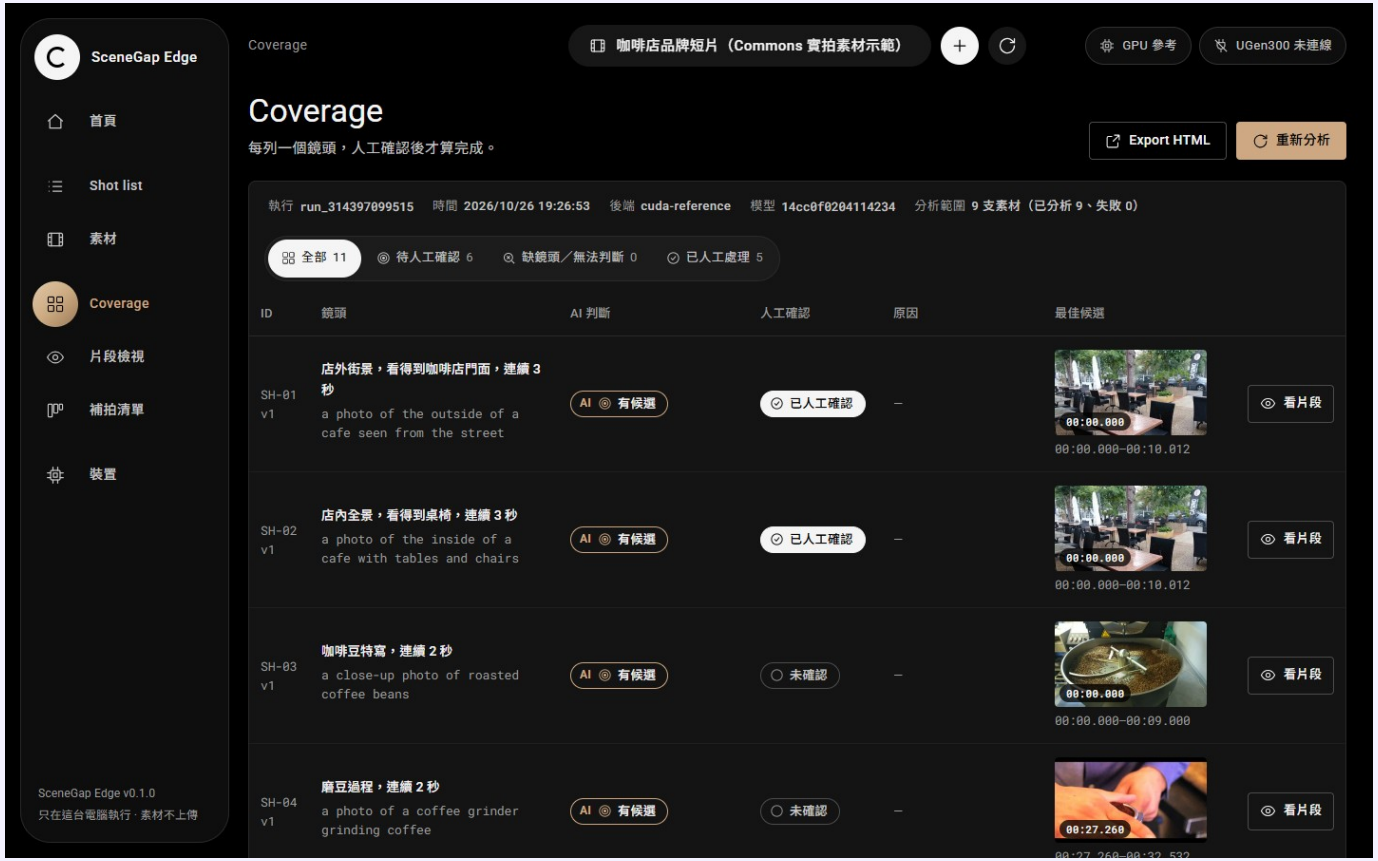
Bound to the requirement revision. Edit the requirement and the old acceptance no longer counts.

Outdated

Requirement, footage, model or threshold changed since the run. The old verdict is hidden, not reused.

SOLUTION

The coverage board a crew reads before leaving



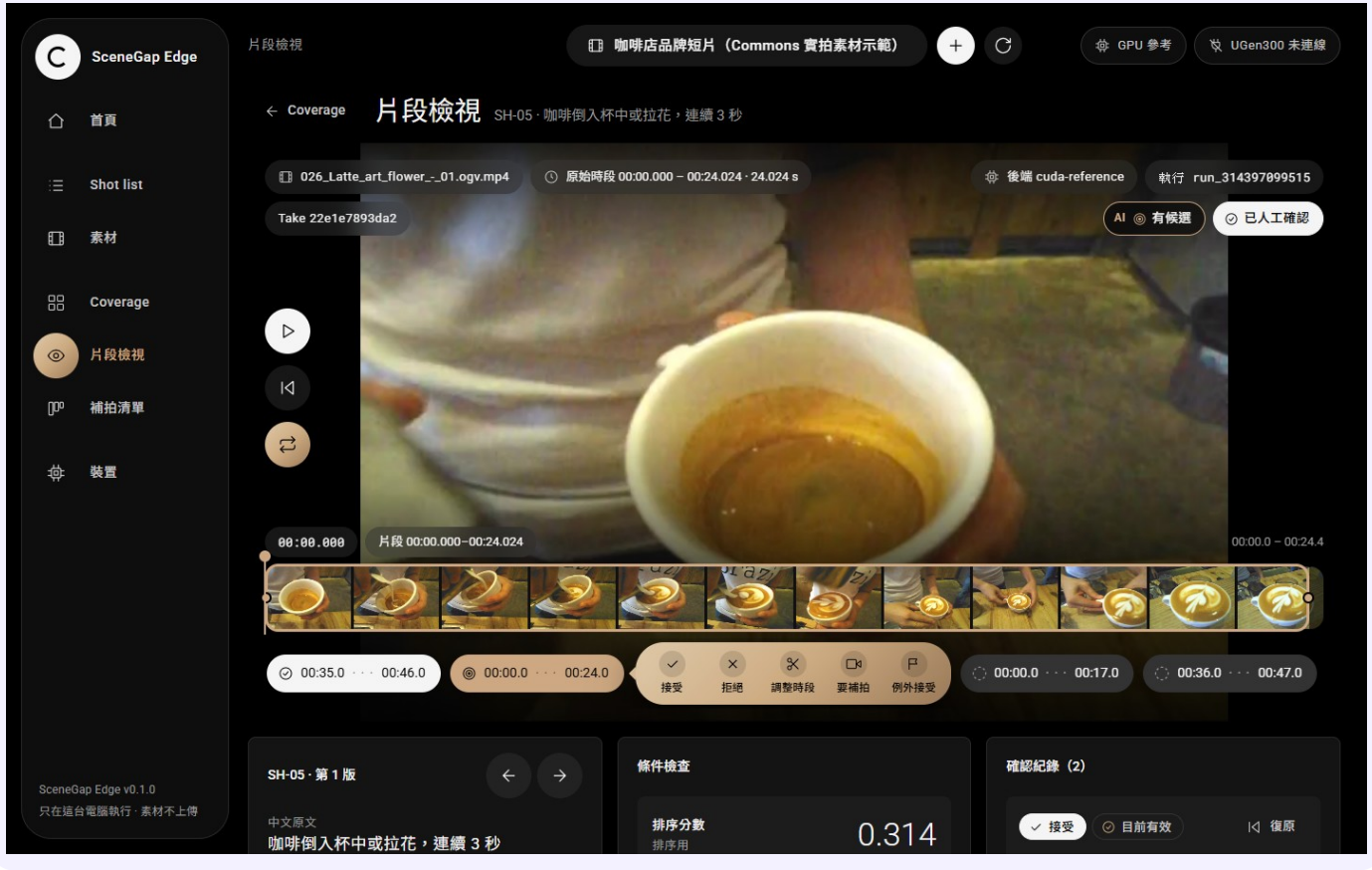
Live app on 9 Wikimedia Commons clips after a CLIP run and a short human review. Frames are real footage.

Read in one pass

- Each row: the shot, the AI's answer, the person's answer, the best clip
- Coverage counts only shots a person confirmed
- Filters: waiting for review, missing, done

SOLUTION

Every verdict plays its evidence



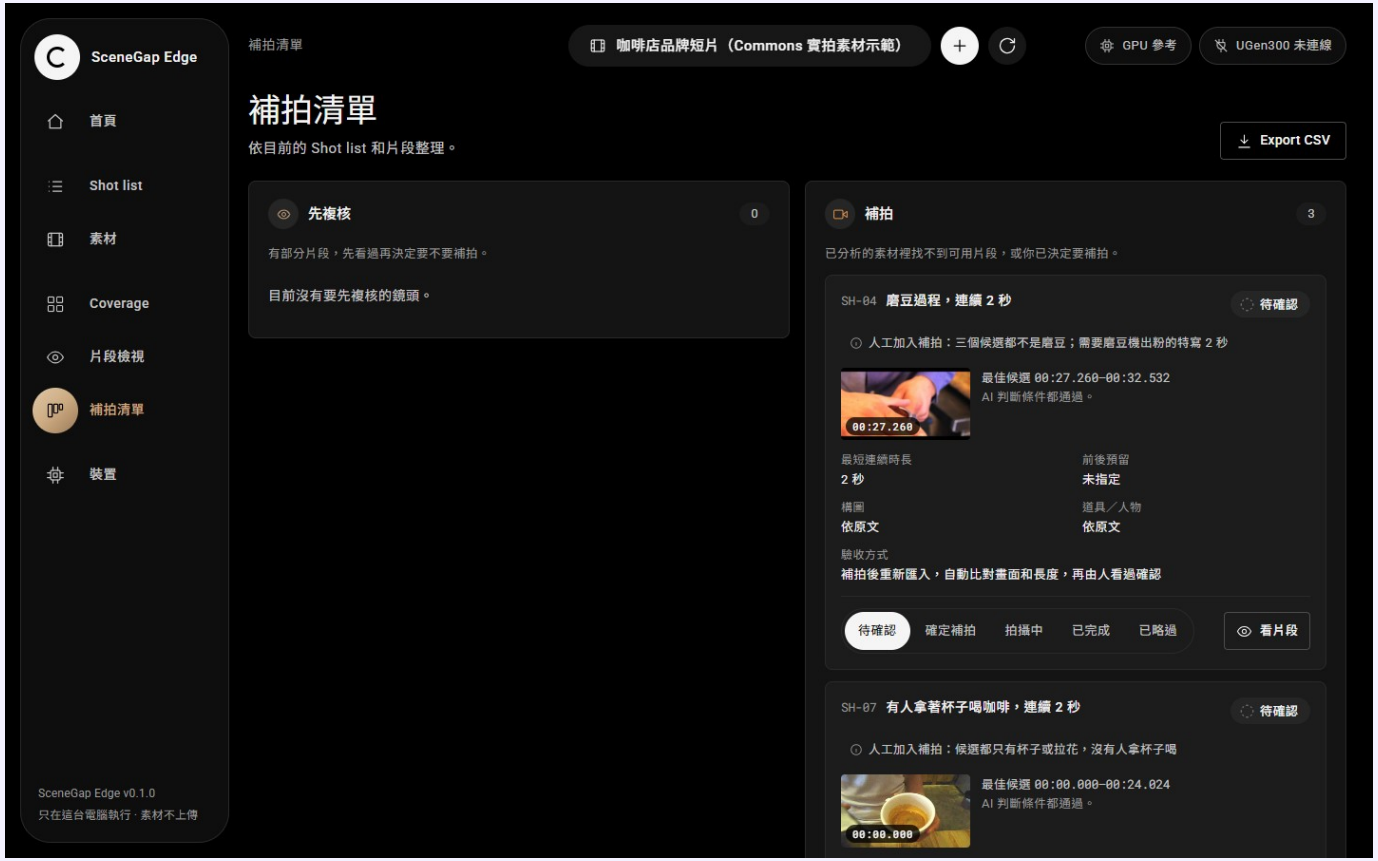
Latte-art clip accepted for 'coffee poured into a cup'; the espresso-only clip before it was rejected (no cup in frame).

One click per decision

- Accept, reject, adjust range, send to reshoot
- Filmstrip and range chips on the original timecode
- Every decision is logged and can be undone
- Original files are never touched

SOLUTION

Gaps become a reshoot list before the crew leaves



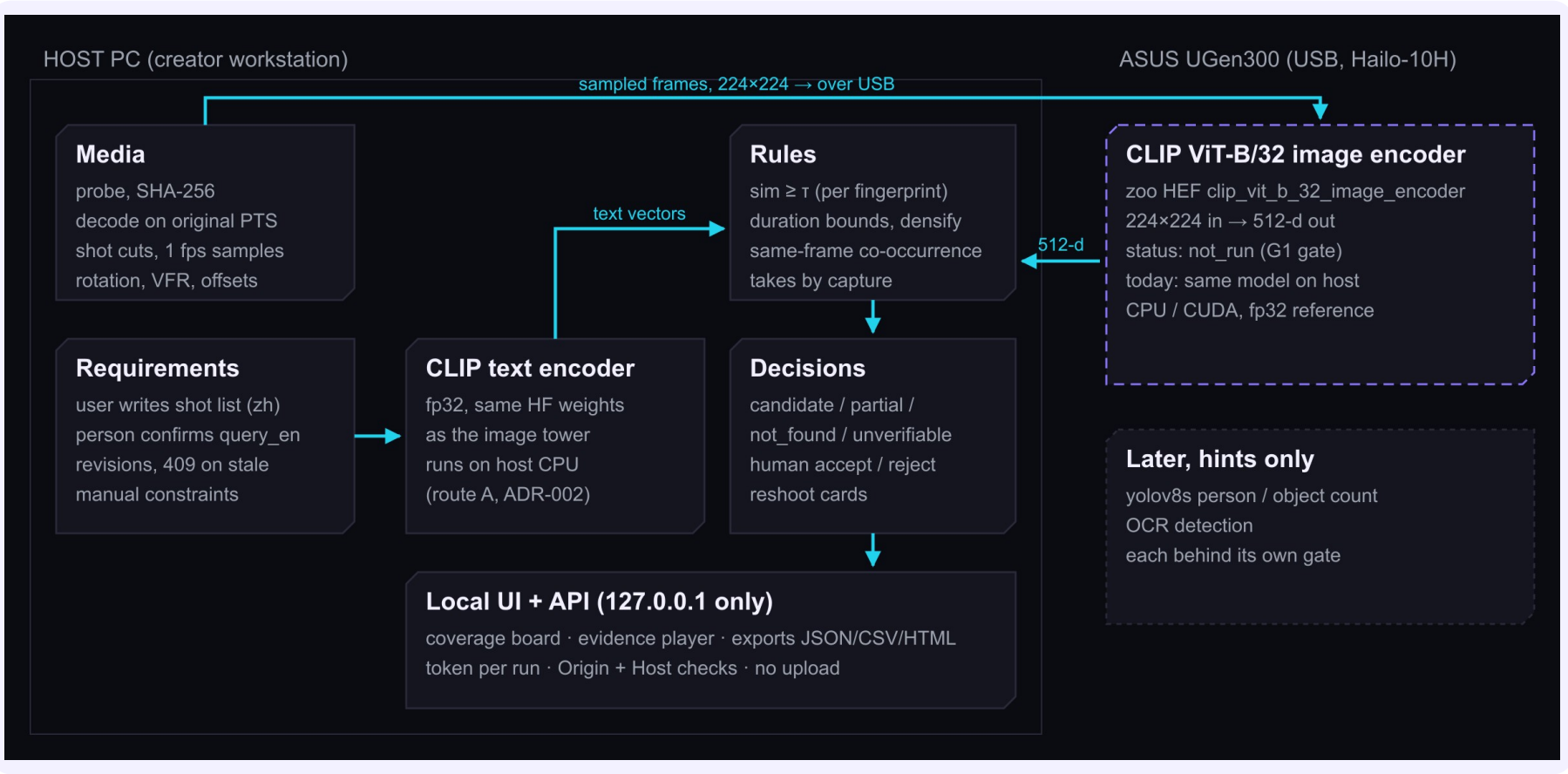
Reshoot list from the same demo: grinder, someone drinking, and the coffee bag were not in the footage.

What the crew gets

- The shot, why it is missing, the closest clip
- Minimum length and how it will be checked
- Status: to confirm → shooting → done

HARDWARE

UGen300 takes the image encoder; the host keeps the rest



Route A (ADR-002): CLIP ViT-B/32 image tower as Hailo's zoo HEF; text tower on the host with the same weights.

HARDWARE

UGen300 plan: image encoder on the device, checked by a G1 gate

GATE	MEASURE (DEV FRAMES, SAME PREPROCESSING)	PASS BAR	STATUS
G1a	Cosine, HEF vs fp32 embedding, per frame	median ≥ 0.98, p5 ≥ 0.95	not_run
G1b	Spearman ρ of similarity per query	median ≥ 0.95, worst ≥ 0.85	not_run
G1c	Top-5 / top-10 shot overlap	≥ 0.80 / ≥ 0.85	not_run
G1d	Dev Recall@5, each backend with its own τ	≥ fp32 − 3 pts	not_run
G1f	Frames/s over USB, 60 s clip end to end	reported, no bar	not_run
Harness	Same script, CPU vs CUDA fp32 (self-check)	≈ 1.0 expected	cos 1.0000, ρ 1.000

> **Vendor: Hailo-10H ViT-B/32 image encoder, CIFAR-100 zero-shot 65.3 float → 64.5 on chip (PCIe, not USB).**

docs/MODEL_PLAN.md §2, §4; Hailo Model Zoo (vendor figures).

EVIDENCE

How we tested without fooling ourselves

Step 1

Protocol first

Metrics, split, threshold rule committed to git before any label existed.

Step 2

Requirements

11 café-promo requirements and queries written before any frame was seen.

Step 3

Two annotators

Independent labels on 1 fps contact sheets; a third pass on every disagreement.

Step 4

Dev only

Threshold τ and crop policy chosen on dev clips, then frozen.

Step 5

Test once

Held-out uploaders, scored once; the script refuses a second run.

EVIDENCE

Real footage, open licences, not shot by us

44

Wikimedia Commons clips, 25 min, every licence and SHA-256 in the manifest

Licences: CC BY-SA 4.0 ×15, CC BY 3.0 ×9, CC0 ×8, CC BY-SA 3.0 ×4, PD ×3, CC BY 4.0 ×3, CC BY 2.0 ×2

0.91 Cohen's κ between the two annotators (sample level)

26 / 18 test / dev clips, split by uploader so no shoot is on both sides

11 requirements, incl. a co-occurrence and a manual brand check

EVIDENCE

On held-out footage, the right clip is in the top 5 for 7 of 11 shots

METHOD (SAME CLIP, SAME T)	RECALL@5	PRECISION@5
Title keyword match	45% (5/11, 95% CI 21–72%)	36% (8/22, 95% CI 20–57%)
Single-frame semantic search	73% (8/11, 95% CI 43–90%)	40% (22/55, 95% CI 28–53%)
SceneGap: + duration, densify, same frame	**64% (7/11, 95% CI 35–85%)**	**33% (18/55, 95% CI 22–46%)**

> **26 held-out clips, scored once at 2026-10-26 19:22:51 with τ = 0.225 (center-crop).**

eval/results/test_metrics.json; Wilson 95% intervals.

EVIDENCE

Finding missing shots automatically is the open problem

METHOD	GAP RECALL	WRONG RESHOOT	UNKNOWN	CANDIDATE WHERE NONE EXISTS
Title keyword match	89%	44%	0%	11%
Single-frame semantic	0%	0%	0%	100%
SceneGap	**0%**	**0%**	**12%**	**79%**

> **The threshold was tuned for single frames, so almost every shot finds some candidate. On dev, a margin score finds 33% of gaps at 11% wrong reshoots; Stage II calibrates per shot.**

eval/results/test_metrics.json; explore_dev_gap_tradeoff.json (dev only).

EVIDENCE

Analysis costs 2–3 s per minute of video; decoding, not the model, dominates

BACKEND	DEVICE	ANALYSE (DECODE + EMBED)	SEARCH P50 / P95, 10K SEGMENTS	MODEL LOAD
cpu-reference	desktop CPU	3.1 s per minute of video	11.77 / 14.29 ms	6.51 s
cuda-reference	NVIDIA GeForce RTX 4080	2.3 s per minute of video	7.61 / 13.39 ms	1.88 s
UGen300	Hailo-10H	not_run	not_run	not_run

1080p30 playback while analysing: 0 late frames; frame work p95 none 5.72 ms / cpu 7.54 ms / cuda 6.24 ms (budget 33 ms).

eval/results/perf.json, interference.json.

ENGINEERING

Built to be checked: tests, trace, safety

46 automated tests; 30 of the PRD's 36 acceptance cases pass by test; a fresh-environment install passes the suite.

Media edge cases

Rotation, 29.97 fps, VFR, start offsets, corrupt and unsupported codecs, files still copying, moved files found again by hash.

Logic

Three-valued AND/OR/NOT; two short pieces never add up; duplicates are not two takes; revisions void old decisions (409 on stale edits).

Security

127.0.0.1 only, per-run token, Origin + Host checks, authorised folders, CSV formula guard, escaped HTML, no traceback to the browser.

Offline

Core flow tested with outbound sockets blocked. Footage never leaves the computer.

Failure

Disk full stops safely and resumes; interrupted jobs resume; originals' hashes unchanged.

Rebuild

Fresh venv from the public index passes the suite; Dockerfile and CI included; Apache-2.0.

BUSINESS

Who pays: small crews and freelance editors who shoot on location

SIGNAL	NUMBER	SOURCE
US film and video editors / camera operators (2025)	39,400 / 32,600; 35% self-employed	US BLS OOH
Median pay, editors (May 2025)	$75,420 / yr	US BLS OOH
Day rate, small crew (estimate)	NT$12,000–28,000 (TW); $800–1,500 (US)	PRO360 quotes; rate cards
Creators worldwide	~50M, ~4% professional	Goldman Sachs (partial)
Shot-list practice	crews tick shots to schedule pickups	StudioBinder guide

> **Value per project ≈ review hours saved × rate + avoided reshoot day − setup and review time. Unproven until the Stage II user study.**

docs/research/BUSINESS_FACTS.md (URLs, access dates, verified flags).

BUSINESS

Pricing hypothesis: one-time licence, sold with the accelerator

All prices here are hypotheses to test in interviews (PRD 22.3), not a market claim. ASUS has not endorsed this product.

Perpetual licence

Local-first users dislike per-minute upload pricing. Benchmark: Resolve Studio $295, Final Cut $299.99 one-time.

With UGen300

UGen300 USB reported at $299.99 (CNX Software). A bundle adds a concrete creator workflow to a USB accelerator.

Template packs

Reusable shot-list templates per genre (café, product, event) with tuned queries; yearly updates.

PLAN

Stage II: device, users, Chinese queries

Wk 1–2

G1 on a UGen300

Run the HEF image encoder; pass G1a–d or report the failure and stay on cpu-reference.

Wk 2–3

Speed with the device

Same clips, CPU vs GPU vs NPU; playback interference in a real NLE timeline.

Wk 3–5

User study

3–5 interviews, then 8–12 crews: manual vs SceneGap, counterbalanced; time, misses, wrong reshoots.

Wk 5–6

Chinese queries

SigLIP2 multilingual on device; adopt only if within 3 pts of English Recall@5 on dev.

Wk 6+

Hints

yolov8s person/object counts and OCR detection, each behind its own gate.

PLAN

What could break, and what we do then

RISK	FALLBACK
HEF fails G1 (quantisation hurts ranking)	Recalibrate with dev frames; TinyCLIP; else ship cpu-reference and say so
False cut splits a long take → false 'too short'	Near-threshold cuts give UNKNOWN, not FAIL; the cut is shown in the viewer
One global τ fits some queries badly	τ per model fingerprint now; per query family next, calibrated on dev
English query drifts from the Chinese requirement	Person confirms query_en; multilingual model only after its gate
AI annotators, Commons footage, small test split	Wide CIs shown; Stage II adds human labels and self-shot projects
Crews will not write shot lists	Templates; the interview checks this first

Fewer surprises in the edit

SceneGap Edge · local-first shot coverage for small crews

- Checks the shot list, not just the footage: length, takes, same frame
- Finds the right clip in the top 5 for 7 of 11 shots on held-out footage; a person confirms in one click
- Next: the image encoder on UGen300, per-shot thresholds for missing-shot detection, and a user study

APPENDIX

Footage credits (1 of 3)

#	WIKIMEDIA COMMONS FILE	AUTHOR	LICENCE
000	Cadaques cafe de la habana 2008 august converted YAN3.ogv	Cliché Online (talk) (Yan.)	CC BY 3.0
001	Vichy - Grand Café, terrasse sous la grêle.ogv	Romainbehar	CC0
002	Merry X-mas at Kranzler.webm	This video is a work by Wikipedia and Wi	CC BY-SA 4.0
003	Times Cafe inflatable waving character.webm	Andy Li	CC0
004	Rateel cafe music2.webm	Iainsimpsonstewart	CC0
005	Starbucks en la aĉetcentro Kristall (Tjumeno).webm	RG72	CC BY-SA 4.0
006	Interior Paragon Café Katoomba.ogv	Bluedawe	CC BY-SA 3.0
007	Café Lomi, Paris - Vimeo.webm	ideafarm films	CC BY 3.0
008	Operation Ghost Stories - Video 01.webm	FBI – Federal Bureau of Investigation	Public domain
009	Coffee Bean & Tea Leaf Process.webm	Kelliks	CC BY-SA 3.0
010	“走线客”过境墨西哥 不忘来杯星巴克 .webm	美国之音中文网	Public domain
011	Color sorter coffeebeans.webm	AlexanderKlink	CC0
012	Kaffe - Ystad-2024.ogg	Foto: Jonn Leffmann	CC BY 3.0
013	Coffee from dan 2.webm	Denis kasozi	CC BY-SA 4.0
014	Traditional Ethiopian Coffee Roasting Ceremony.webm	Keria T	CC BY-SA 4.0

APPENDIX

Footage credits (2 of 3)

#	WIKIMEDIA COMMONS FILE	AUTHOR	LICENCE
015	Despulpadora.ogg	Jhonespe150102	CC BY-SA 4.0
016	Lions Almelo zamelt koffie in voor Voedselbank.webm	AAVISIE	CC BY 3.0
017	Sunday Morning Espresso Routine - Grind + distribute + tamp	Scott Schiller from San Francisco, CA, U	CC BY 2.0
018	Кофемельница JURA 43.webm	Gontcharik.petr	CC BY-SA 4.0
019	MoliendoCafeEnLicuadora ohs02.webm	PetrohsW	CC BY-SA 4.0
020	Espresso video.ogv	Rosatrieu	CC BY-SA 3.0
021	Macro Espresso Process.webm	Scott Schiller from San Francisco, CA, U	CC BY 2.0
022	Der perfekte Espresso aus dem Siebträger.ogv	Max Duchardt, Katrin Engel	CC BY 3.0
023	Pour Over.ogv	Lev Lazinskiy	CC BY-SA 4.0
024	Use of a coffee press.ogv	Produced, directed and edited by Daniel	CC BY-SA 3.0
025	Schritt für Schritt - Kaffeezubereitung mit der Chemex.ogv	Max Duchardt , Katrin Engel	CC BY 3.0
026	Latte art flower - 01.ogv	Anna Frodesiak	CC0
027	Latte art leaf - 01.ogv	Anna Frodesiak	CC0
028	Nespresso machine.ogv	Kjetil Ree	CC0
029	A man pours Arabic coffee.ogv	Sarah Canbel	CC BY-SA 4.0

APPENDIX

Footage credits (3 of 3)

#	WIKIMEDIA COMMONS FILE	AUTHOR	LICENCE
030	Servido tradicional de café etíope.webm	ProtoplasmaKid	CC BY 4.0
031	How to brew Coffee using the laws of physics - with the Hari	Coffee Circle	CC BY 3.0
032	Sugar falling into coffee.webm	Anselm Schüler	CC BY-SA 4.0
033	Bosnian coffee with rahat lokum, Kujundziluk and Old Bridge	Minja Klarić	CC BY-SA 4.0
034	A cup of Kenyan Coffee.webm	Queen Asali	CC BY-SA 4.0
035	Latte salute.webm	Unidentified. Released by the White Hous	Public domain
036	Turkish tea 2015.webm	Klapi	CC BY-SA 4.0
037	Jane pouring beer fast.webm	Angulidayaaluta	CC BY-SA 4.0
038	Turkovich Family Wines - June 2019 - Stierch 03.ogv	Missvain	CC BY 4.0
039	Drinking glass symphony.ogv	IMG 4512	CC0
040	Thermochromic mug.webm	The wub	CC BY-SA 4.0
041	Empty Streets of Paris COVID19.webm	CreatorsCloud	CC BY 3.0
042	IrishTraditionalMusic-IrishPub-Tokyo-2016-2-24.webm	Nesnad	CC BY 4.0
043	Couple holding hands.webm	https://www.youtube.com/user/Editor	CC BY 3.0